ZTP over UDP: Test Report

Version	1.1 (M7 release, after the final review)
Owner	Sunil Vallamkonda
Code under test	commit `0b2e460` (release regression); later commits change only tests, scripts and documents
Test plan	QA_TEST_PLAN.md: 165 cases (§5), impairment matrix (§6), traceability (§10)
Evidence	evidence/release-20261112-220052/ (the release run), evidence/ci/ (CI reports), logs/ (committed logs, logs/README.md)
Date	2026-11-13

This is the release test report that QA §11 asks for: the result of every test case, the impairment matrix with times and retransmission ratios, the performance numbers, and what is open or waived.

1. Summary

Result	Cases	Meaning
✅ PASS	139	Run and passed in the release regression (and in CI where noted)
⚠ Partial	10	Run and passed, except one part: either it needs the open test tooling (QA §12.5), or one platform or sub-case was not run (each row says which)
⚠ Open	15	Logic implemented, not exercised end to end: each needs a hostile-traffic tool that remains with the project owner (PRD R-15, QA §12.5)
⚠ Waived	1	TC-SIM-04 (macOS dummynet, needs `sudo`); TC-SIM-03 (netem in CI) and the in-code simulator cover the same behaviour
❌ Fail	0	
Total	**165**	

Most visible gap. The PDF's example dump has a failed row (ZTP-84BC19, "Failed / Invalid HMAC"). The server records that verdict (DESIGN §10.3), but no committed run shows it: the device needs a wrong key, made with the `--bad-key` test hook, which was not built (TC-AUTH-03 Open; TC-PDF-02, TC-SRV-01, TC-SRV-05, TC-E2E-02 and TC-DEL-02 Partial). The demo runs four of the PDF's five devices.

Runs behind these results

Run	Where	Result
`make release` on `0b2e460`, 2026-11-12 22:00 → 2026-11-13 00:39 PST	Development Mac (Darwin arm64, Apple clang 16.0.0), no sleep during the run	24/24 steps PASS; ASan build 16/16 (summary.txt)
CI run on `0b2e460` (push)	GitHub Actions: Ubuntu gcc and clang, macOS clang, ASan on both, platform, matrix, netem, valgrind	12/12 jobs green (run 43375735275); the 10.10.10.5 Appendix A logs from it are in logs/appendix_a/linux-10.10.10.5/
CI run 52 on `23757cd` (manual, nightly seed counts)	The same jobs, 10-seed matrix, TC-PERF-01/02, netem × 10	12/12 green (report); referenced below for the CI-only cases (the device and server code is that of 0b2e460)

Run	Where	Result
Fresh clone of `6903ab4` (the code of 0b2e460)	Development Mac	`make`: no warnings; `make demo`: PASS (TC-DEL-01, TC-DEL-04)

2. Impairment matrix (QA §6)

The demo (four devices) over each row, seeds 1–10, release run. "Retransmissions" is the share of all datagrams sent (server and devices, every run of the row) that were retransmissions; it and the out-of-order and duplicate counts are totals of the processes' `Counters:` lines (matrix_counters.txt has the table and the command; per-run checks and dumps are in `run_matrix/<row>.<seed>/`, and one run is kept whole in `run_matrix/M3.1/logs/`).

Row	Impairment (each direction)	Runs	Devices OPERATIONAL	Median	p95	Retransmissions	Out of order / dup received	Result
M1	none	10/10	40/40	0.3 s	0.4 s	0.0%	0 / 0	PASS
M2	10% loss, 5% reorder, 2% dup, 1 ms	10/10	40/40	1.6 s	4.0 s	12.6%	730 / 243	PASS
M3	15% loss, 10% reorder, 5% dup, 2±1 ms (the demo)	10/10	40/40	2.7 s	6.3 s	16.6%	1000 / 378	PASS
M4	20% loss, 10% reorder, 5% dup, 5±2 ms	10/10	40/40	4.3 s	9.7 s	19.8%	1017 / 372	PASS
M5	30% loss, 20% reorder, 10% dup (stress, ≥ 95% required)	10/10	40/40	11.0 s	35.4 s	27.4%	1076 / 596	PASS
M6	50% loss	not run						not part of the release run
M7	15% loss correlated 25%, 10% reorder, 5% dup, 5 ms (kernel netem, CI Ubuntu)	10/10 (CI run 52)	all					PASS
M8	20% loss device → server only	10/10	40/40	1.4 s	5.6 s	14.6%	26 / 296	PASS
M9	20% loss server → device only	10/10	40/40	1.6 s	5.7 s	13.7%	760 / 124	PASS

In CI (run 52) rows M1–M4 × 10 seeds passed on Ubuntu and macOS with every device OPERATIONAL. Every run also checks the invariants: no duplicate DB rows, no second REGISTER after a restart, and every artifact hash of an OPERATIONAL device verified (`tests/check_invariants.sh`). The PMTU black-hole detector re-requested chunks in 8 of the 360 device runs of the matrix and perf02 and never stepped down falsely (DESIGN §5.8).

3. Performance (QA §8)

Case	Measure	Target	Result
TC-PERF-01	50 devices at 15% loss	all OPERATIONAL ≤ 120 s, server RSS ≤ 64 MB	50/50 in 21 s, peak RSS 9 MB (Mac); 17 s, 5 MB (CI macOS) and 24 s, 8 MB (CI Ubuntu), run 52

Case	Measure	Target	Result
TC-PERF-02	Time to OPERATIONAL, row M4 × 20	median ≤ 60 s, p95 ≤ 90 s	median 3.4 s, p95 8.7 s (Mac); 3.7 s / 9.6 s (CI macOS) and 3.9 s / 10.5 s (CI Ubuntu), run 52
TC-PERF-03	Transmissions per frame at 10 / 20 / 30% loss	within 10% of the model 1.23 / 1.56 / 2.04	1.16 / 1.36 / 1.76 (m6_perf03.out)
TC-PERF-04	1 h soak, 5 devices	no RSS growth, no leaks	every process's RSS lower at the end than after warm-up (server 1024 → 928 kB); `leaks` : 0 in all 6 processes; 9155 keepalives, 0 session timeouts; valgrind (CI, 3 min): clean
TC-NAT-07	30 min at 20% loss, production timers	0 session timeouts	0 timeouts, 87 keepalives, 1 REGISTER
TC-NAT-13	10 min at 20% loss, demo timers	timeouts recover without re-registration	1 timeout, recovered in 0.3 s, no re-registration

4. Defects found by the release runs and the final review

Found by	Defect	Fix
Release run on 3f9a06c, matrix M3 seed 3	Device: a false PMTU black-hole step-down to 508 B under ordinary loss	821d134: re-request once, step down only after a second silent interval (DESIGN §5.8)
Release run on 821d134	Device: full-size chunks buffered out of order were not counted as data	a1c5bd3: any authenticated CHUNK_DATA over 508 B restarts the interval
Release runs on 3f9a06c and a1c5bd3	Test harness: S4 rebinding check, TC-LOG-02 on loss-only rows, shared suite folders; the Mac slept during one run	821d134, b936db3 (the run fails itself if `pmset` logged a sleep)
CI on dc3c139	Test: TC-SRV-09 counted too few log lines to see interleaving	cfcb3fa
Final review (code)	Server: `limits.max_sessions` above 1536 could hang the session table; an Invalid HMAC REGISTER discarded the half-open session (a forged frame could abort a genuine registration); the rate limit was spent before the MAC check; NAK resends were unbounded; a device with a smaller datagram limit was stepped down to 508	0b2e460 (DESIGN §10.3, §5.8)
Final review (code)	DB journal: a torn last line could swallow the next record; non-ASCII strings made lines unreplayable; write/fsync errors were ignored before the ACK; the session history grew forever	0b2e460 (DESIGN §14.4)
Final review (code)	Device: the forced re-key at 2^{30} (DESIGN §8.9) was not implemented; a crash between the rename and the state save cost a hash attempt; a rollback could be reported as an upgrade after a crash; a queued probe counted as acknowledged	0b2e460
Final review (tests)	TC-UNIT-06 compared a constant with itself; weak checks in TC-UPG-02/03, TC-DISC-02, the `sha()` fallback and the GNU `nm` pattern	0b2e460

One review finding is accepted as a residual risk: a replayed resume HELLO just after a server restart can stall a session briefly (DESIGN §18.3, item 5).

5. Results per test case

Each row: the case from QA §5, its result, where it ran, and the output that shows it. Suites marked "(Mac; CI build-test ×3, asan ×2)" are part of `make test`: they ran in the release run, in its ASan build, and in every CI run on Ubuntu (gcc, clang) and macOS, with and without AddressSanitizer. Each `.out` file is the suite's full output; a suite prints every failed check (`FAIL: TC-…`) and ends with its passed and failed counts, so a PASS row means none of that case's checks failed. To rerun one suite or case: README §16.3.

Unit (L1)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-UNIT-01	SHA-256 known-answer tests	NFR-01, FR-18	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-02	HMAC-SHA256 known-answer tests	FR-13	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-03	Header and TLV encode/decode round trip	FR-05, BON-02	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-04	TLV parser bounds	NFR-07	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-05	Constant-time compare	NFR-06	P2	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-06	Serial-number arithmetic and wraparound	FR-05	P1	✅ PASS	`unit`, `m5_upgrade` (Mac; CI build-test ×3, asan ×2)	unit.out, m5_upgrade.out
TC-UNIT-07	RTO estimator, Karn, backoff	FR-06	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-08	Reorder buffer: in-order delivery	BON-04	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-09	Duplicate classification	FR-08, BON-05	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-10	Generating and parsing SACK	FR-07	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-11	Policy parser and precedence	FR-29	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-12	Version comparison	FR-21	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-13	Atomic file write	NFR-10	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-14	Persisting and loading the chunk bitmap	FR-19	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-15	Netsim determinism and rates	NFR-09, BON-06	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-16	Key-derivation known-answer test	FR-13	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-17	Config key validation	FR-24	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-18	Strict JSON parser	FR-34, NFR-07	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-UNIT-19	Bitmap conversion on chunk-size change	FR-19, FR-20	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-UNIT-20	Precedence and policy validation	FR-34, FR-35	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out

Discovery (DISC)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-DISC-01	Soft serial generated on first boot	FR-01	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2); `scenario_logs` (Mac; logs/scenarios/)	m2_session.out, scenario_logs.out
TC-DISC-02	Soft serial loaded on later boots, and assigned by configuration	FR-01	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2); `scenario_logs` (Mac; logs/scenarios/)	m2_session.out, scenario_logs.out
TC-DISC-03	Static IP discovery	FR-02	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-DISC-04	Multicast discovery	BON-01	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-DISC-05	Broadcast discovery	BON-01	P2	⚠ Partial	`m2_session` (Mac; CI build-test ×3, asan ×2); CI `build-test` (Ubuntu): broadcast, part (b)	m2_session.out, CI run 52 — part (b), broadcast on Ubuntu CI, runs; part (a), broadcast on the Mac, is skipped (no BCAST address set locally)
TC-DISC-06	Fallback to static	BON-01	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-DISC-07	Forged OFFER ignored	BON-01, NFR-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-DISC-08	Replayed OFFER ignored	BON-01	P2	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out

Session (SESS)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SESS-01	Handshake happy path	FR-04	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-SESS-02	HELLO lost: idempotent retransmission	FR-06, BON-05	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SESS-03	HELLO_ACK lost	FR-06, BON-05	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-SESS-04	Session IDs are random and unique	FR-04	P2	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-SESS-05	Server down: retry exhaustion and recovery	FR-06, FR-11	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-SESS-06	Unknown session_id dropped	NFR-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-SESS-07	Stale epoch dropped	FR-23	P1	⚠ Open	—	QA §12.5; needs an old-epoch frame injector
TC-SESS-08	UDP only	FR-03, DEL-04	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-SESS-09	A HELLO can't tear down a live session	FR-14, NFR-06	P1	⚠ Open	—	QA §12.5; needs a HELLO takeover tool

Reliability (REL)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-REL-01	Every frame carries a seq	FR-05, BON-02	P1	✅ PASS	CI `platform-linux` : packet capture checked frame by frame	CI run 52
TC-REL-02	Retransmission on timeout	FR-06, FR-07, BON-03	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-REL-03	The PDF scenario, reproduced verbatim: seq 2 lost, seq 3 buffered, duplicate of seq 2	BON-04, BON-08	P1	✅ PASS	`unit` , `m2_peer` (Mac; CI build-test ×3, asan ×2)	unit.out, m2_peer.out
TC-REL-04	Lost ACK: data received	BON-07, BON-05	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-REL-05	Heavy duplication	FR-08, BON-05	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-REL-06	Window limit respected	FR-06	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-REL-07	Fast retransmit	FR-06	P2	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-REL-08	Out-of-window frames dropped	NFR-07	P2	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-REL-09	Completes at 10/15/20% loss	FR-09, NFR-03	P1	✅ PASS	`matrix` (Mac: M1–M5, M8, M9 × 10); CI `impair-matrix` (Ubuntu, macOS)	matrix.out, CI run 52
TC-REL-10	Heavy reordering	BON-04	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-REL-11	No ACK loops	FR-07	P2	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out

NAT and keepalive (NAT)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-NAT-01	Keepalives when idle	FR-10	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-02	No keepalives during a transfer	FR-10	P2	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-03	Device detects a dead server	FR-11	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2); `scenario_logs` (Mac; logs/scenarios/)	m5_gate.out, scenario_logs.out
TC-NAT-04	Server detects an idle device	FR-11	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-05	NAT rebinding	FR-11, FR-23	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-NAT-06	Spoofed source can't hijack	NFR-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-NAT-07	No false timeouts under loss (production timers)	FR-11, NFR-03	P2	✅ PASS	`longrun` (Mac)	longrun.out
TC-NAT-13	False timeouts with demo timers recover cleanly	FR-11, FR-23	P2	✅ PASS	`longrun` (Mac)	longrun.out
TC-NAT-08	NAT idle timeout shorter than the keepalive	FR-36, FR-35, FR-11	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-09	NAT reboot: all mappings flushed	FR-36, FR-23	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-NAT-10	Public IP change	FR-36	P2	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2); CI `build-test` (Ubuntu): strict-IP resume needs 127.0.0.2, which macOS lacks (the Mac run prints "skipped")	m5_upgrade.out, CI run 52

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-NAT-11	Outbound UDP blocked by a customer firewall	FR-36, A-09	P1	✅ PASS	`m2_session` (Mac; CI build-test ×3, asan ×2)	m2_session.out
TC-NAT-12	A replayed valid frame from a new address can't move the peer	NFR-06, FR-14, FR-11	P1	⚠ Partial	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out — KEEPALIVE half passes; the CHUNK_REQ replay half needs a replay tool

Authentication and registration (AUTH)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-AUTH-01	REGISTER contents	FR-12	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-AUTH-02	Valid registration	FR-13	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-AUTH-03	Invalid HMAC (wrong model key)	FR-13, FR-30	P1	⚠ Open	—	QA §12.5; needs `--bad-key`
TC-AUTH-04	Replay in the same session	FR-14	P1	⚠ Open	—	QA §12.5; needs `--replay-test`
TC-AUTH-05	Replay spliced into a new session	FR-14	P1	⚠ Open	—	QA §12.5; needs `--replay-test`
TC-AUTH-06	Reused nonce	FR-14	P1	⚠ Open	—	QA §12.5; needs `--reuse-nonce`
TC-AUTH-07	Replayed resume HELLO	FR-14, FR-23	P1	⚠ Open	—	QA §12.5; needs a resume replay tool
TC-AUTH-08	Tampering with any byte	NFR-06	P1	⚠ Open	—	QA §12.5; needs a bit-flip injector (TC-UNIT-03 flips every bit of a frame)
TC-AUTH-09	Unknown model	FR-29, FR-13	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out
TC-AUTH-10	HELLO flood is rate-limited	NFR-05	P2	⚠ Open	—	QA §12.5; needs a HELLO flood tool
TC-AUTH-11	Rogue server rejected	NFR-06	P1	⚠ Open	—	QA §12.5; needs a rogue-server `peer_sim` mode
TC-AUTH-12	Known model, no policy	FR-29, FR-13	P1	✅ PASS	`m3_register` (Mac; CI build-test ×3, asan ×2)	m3_register.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-AUTH-13	Wrong fleet key (not a fleet device)	NFR-06, FR-36	P2	⚠ Open	—	QA §12.5; needs a wrong-fleet-key sender
TC-AUTH-14	Forged or replayed NAK is ignored	FR-14, NFR-06	P1	⚠ Open	—	QA §12.5; needs forged/replayed NAKs
TC-AUTH-15	Replayed terminal ERROR is ignored	FR-14, NFR-06	P1	⚠ Open	—	QA §12.5; needs replayed terminal errors

Provisioning (PROV)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PROV-01	Config in chunks with hash	FR-15, FR-17, FR-18	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-02	Config is valid JSON	FR-15	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-03	Firmware in chunks with hash	FR-16, FR-17, FR-18	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-04	Datagram size cap	FR-20, NFR-02	P1	✅ PASS	CI `platform-linux`: packet capture	CI run 52
TC-PROV-05	DF set, no IP fragments	FR-20	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2); CI `platform-linux`: packet capture	unit.out, CI run 52
TC-PROV-06	Chunk boundaries	FR-17	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-07	Hash mismatch handling	FR-18	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-08	Resume after kill -9	FR-19	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-09	Resume after a server restart	FR-19, FR-23	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-PROV-10	Artifact changed between attempts	FR-18, FR-19	P2	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out
TC-PROV-11	Duplicate chunks are harmless	BON-05	P1	✅ PASS	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PROV-12	1280-byte MTU path: no fallback needed	FR-20, NFR-02	P1	✅ PASS	CI `platform-linux`: loopback MTU 1280	CI run 52
TC-PROV-13	Small-MTU path: step-down to 508 B	FR-20, FR-19	P1	⚠ Partial	`m4_provision` (Mac; CI build-test ×3, asan ×2)	m4_provision.out — parts (b) and (c) run on the Mac and in CI; part (a), a real 576-byte MTU path, needs a CI job that was not built

Upgrade and restart (UPG)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-UPG-01	Outdated firmware upgraded	FR-21	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-02	Process restart	FR-22	P1	✅ PASS	`m5_gate`, `m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_gate.out, m5_upgrade.out
TC-UPG-03	Resume without re-registration	FR-23	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-04	One DB row per serial	FR-23, FR-30	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-UPG-05	Already current: no upgrade	FR-21	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-06	Legacy device path	FR-29, FR-21	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-UPG-07	Crash at each upgrade step	NFR-10	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-08	Grace period expired before the resume	FR-23	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-09	Rollback on failed validation	FR-25	P2	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-10	Fast restart while the server session is still live	FR-23, FR-19	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-UPG-11	Crash during resume, then resume again (no deadlock)	FR-23, NFR-10	P1	✅ PASS	`m5_gate`, `m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_gate.out, m5_upgrade.out

Validation (VAL)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-VAL-01	Happy-path validation	FR-24..27	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-VAL-02	Config missing at validation	FR-24	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-VAL-03	Firmware version mismatch	FR-25	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out
TC-VAL-04	Unhealthy session	FR-26	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-VAL-05	Server records OPERATIONAL	FR-27, FR-30	P1	✅ PASS	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out

Server (SRV)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SRV-01	5-device concurrent demo	FR-28	P1	⚠ Partial	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out — 4 concurrent devices; the expected dump includes the ZTP-84BC19 row, which needs `--bad-key`
TC-SRV-02	Per-device assignment	FR-29	P1	✅ PASS	`unit`, `m6_server` (Mac; CI build-test ×3, asan ×2)	unit.out, m6_server.out
TC-SRV-03	DB record fields	FR-30	P1	✅ PASS	`unit`, `m3_register` (Mac; CI build-test ×3, asan ×2)	unit.out, m3_register.out
TC-SRV-04	Dump on demand	FR-31, DEL-11	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-SRV-05	Summary	FR-32	P1	⚠ Partial	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out — success count checked; the failure count needs the S2 device (`--bad-key`)
TC-SRV-06	DB persists across a server restart	NFR-10	P1	✅ PASS	`unit`, `m6_demo` (Mac; CI build-test ×3, asan ×2)	unit.out, m6_demo.out
TC-SRV-07	Fuzzing malformed datagrams	NFR-07	P1	✅ PASS	`m6_fuzz` (Mac; CI build-test ×3, asan ×2)	m6_fuzz.out
TC-SRV-08	Session limits and half-open expiry	NFR-05, NFR-07	P2	⚠ Open	—	QA §12.5; needs a half-open flood tool
TC-SRV-09	Interleaved progress	FR-28	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-SRV-10	Resume across a server restart mid-upgrade	FR-23, NFR-10	P1	✅ PASS	`m6_server` (Mac; CI build-test ×3, asan ×2)	m6_server.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SRV-11	Failed-record cap	NFR-07, NFR-05	P2	⚠ Open	—	QA §12.5; needs 20,000 failing registrations (cap unit-tested at 1,000 over 100)

Simulation (SIM)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-SIM-01	Observed loss rate, per direction and end to end	BON-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out
TC-SIM-02	Reordering and duplication are observable	BON-06	P1	✅ PASS	`m2_session`, `m4_provision` (Mac; CI build-test ×3, asan ×2)	m2_session.out, m4_provision.out
TC-SIM-03	The netem alternative	BON-06	P2	✅ PASS	CI `impair-netem`	CI run 52
TC-SIM-04	The macOS dummynet alternative	BON-06	P2	⚠ Waived	—	Waived: macOS dummynet needs `sudo`; TC-SIM-03 and the in-code simulator cover it

Performance (PERF)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PERF-01	50 concurrent devices	NFR-05, FR-28	P1	✅ PASS	`scale` (Mac); CI `impair-matrix` (Ubuntu, macOS)	scale.out, CI run 52
TC-PERF-02	Time to OPERATIONAL	NFR-04	P1	✅ PASS	`matrix` (Mac: M1–M5, M8, M9 × 10); CI `impair-matrix` (Ubuntu, macOS)	matrix.out, perf02.out, CI run 52
TC-PERF-03	Retransmission overhead against the model	FR-09	P3	✅ PASS	`m6_perf03` (Mac; CI build-test ×3, asan ×2)	m6_perf03.out
TC-PERF-04	Soak with no leaks	NFR-07	P2	✅ PASS	`soak` (Mac, 1 h); CI `valgrind`: 3 min under memcheck	soak.out, CI run 52

Logs (LOG)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-LOG-01	Required tags present	FR-33	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-LOG-02	Bonus log lines present in the demo	BON-08	P1	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out
TC-LOG-03	Format and files	NFR-08	P2	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-LOG-04	Counters at exit	NFR-08	P2	✅ PASS	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out

Deliverables (DEL)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-DEL-01	Clean build on both platforms	DEL-02, NFR-01, NFR-11	P1	✅ PASS	Fresh clone of 6903ab4, `make` (0 warnings), and the CI build jobs (Ubuntu gcc and clang, macOS clang; `-Werror`)	the binaries link only the C library: linkage_macos.txt
TC-DEL-02	The demo produces its artifacts	DEL-01, DEL-09, DEL-10, DEL-11	P1	⚠ Partial	`make demo` (Mac, release run)	demo.out, logs/ — 4 of 5 device logs; ZTP-84BC19 needs `--bad-key`
TC-DEL-03	Device/server separation	DEL-03	P1	✅ PASS	`make check-separation` (suite `separation`)	separation.out
TC-DEL-04	The README quick start works	DEL-05	P1	✅ PASS	Fresh clone of 6903ab4, README quick start	`make` and `make demo` PASS
TC-DEL-05	README: 5 required answers	DEL-06	P1	✅ PASS	README §11 (review)	Q1–Q5 present
TC-DEL-06	README: design sections	DEL-07	P1	✅ PASS	README §4–§10 (review)	all 6 design topics present
TC-DEL-07	README: bonus coverage	DEL-08	P1	✅ PASS	README §13 (review)	3 topics and B1–B4 present
TC-DEL-08	Sanitizer-clean demo	NFR-11	P1	✅ PASS	`make asan` (ASan build, `m6_demo`)	asan/m6_demo.out
TC-DEL-09	Logs committed	DEL-09, DEL-10	P1	✅ PASS	`git ls-files logs/`; `check_pdf05`; `check_pdf05` (Mac; CI build-test ×3, asan ×2)	check_pdf05.out, logs/
TC-DEL-10	README: why keepalives matter	DEL-12	P1	✅ PASS	README §12 (review)	keepalive section present

End-to-end (E2E)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-E2E-01	Appendix A flow, single device	FR-33, FR-01..27	P1	✅ PASS	`check_appendix_a` (Mac and CI)	check_appendix_a.out
TC-E2E-02	Full demo under impairment	FR-28..32, NFR-03	P1	⚠ Partial	`matrix` (Mac: M1–M5, M8, M9 × 10); CI `impair-matrix` (Ubuntu, macOS)	matrix.out, CI run 52 — all devices OPERATIONAL in every run; ZTP-71DD08 failed_attempts = 1 needs `--replay-test`

Conformance with PDF Appendix A (APX), one test per step

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-APX-01	Step 1: Device boot (device information: ZTP-93AF21, EDGE-GW-100, v1.0)	FR-01, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-02	Step 2: Server discovery (static IP 10.10.10.5, UDP port 5555)	FR-02, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2); CI `platform-linux` : on 10.10.10.5:5555	check_appendix_a.out, CI run 52
TC-APX-03	Step 3: UDP session (HELLO seq=1 with soft_serial; HELLO_ACK seq=1 with session_id=7421)	FR-03, FR-04, FR-05, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2); CI `platform-linux` : on 10.10.10.5:5555	check_appendix_a.out, CI run 52
TC-APX-04	Step 4: NAT keepalive	FR-10, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-05	Step 5: Secure registration (session_id=7421, soft_serial, model, firmware v1.0, nonce=984312)	FR-12, FR-13, FR-29, FR-33	P1	⚠ Partial	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out — REGISTER fields and order checked; the literal nonce 984312 needs `--nonce`
TC-APX-06	Step 6: Config provisioning (the example config, 3 chunks)	FR-15, FR-17, FR-18, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-07	Step 7: Firmware upgrade (v1.0 → v1.1)	FR-16, FR-21, FR-22, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-08	Step 8: Resume after restart, without duplicate registration	FR-22, FR-23, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-09	Step 9: Validation and operational mode	FR-24..27, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out
TC-APX-10	Steps 1–9 under impairment	FR-09, NFR-03, FR-33	P1	✅ PASS	`check_appendix_a` (Mac; CI build-test ×3, asan ×2)	check_appendix_a.out

Conformance with the PDF's other examples and its scope (PDF)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PDF-01	§1 "Example output"	FR-01, FR-02, FR-33	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-PDF-02	§9 "Example of a Dump from Server DB"	FR-31, FR-30, FR-32, DEL-11	P1	⚠ Partial	`m6_demo` (Mac; CI build-test ×3, asan ×2)	m6_demo.out — the dump shows 4 of the PDF's 5 rows; the ZTP-84BC19 "Invalid HMAC" row needs `--bad-key`

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-PDF-03	Appendix A "Example Configuration"	FR-15	P1	✅ PASS	`m5_gate` (Mac; CI build-test ×3, asan ×2)	m5_gate.out
TC-PDF-04	Appendix A addressing (10.10.10.5:5555)	FR-02	P3	✅ PASS	`check_appendix_a`, `m5_gate` (Mac; CI build-test ×3, asan ×2); CI `platform-linux`: on 10.10.10.5:5555	check_appendix_a.out, m5_gate.out, CI run 52
TC-PDF-05	Overview / Scope & Expectations: UDP only, "not a REST or API exercise", C or Python, no UI	FR-03, NFR-01, DEL-04	P1	✅ PASS	`check_pdf05` (Mac; CI build-test ×3, asan ×2)	check_pdf05.out

Configuration (CFG)

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-CFG-01	Built-in defaults	FR-34	P1	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-02	A config file overrides the defaults	FR-34	P1	✅ PASS	`unit` (Mac; CI build-test ×3, asan ×2)	unit.out
TC-CFG-03	Precedence: defaults < file < profile < CLI	FR-34	P1	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-04	Range and type validation (fail fast)	FR-34, NFR-07	P1	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-05	Unknown keys rejected (typo protection)	FR-34	P1	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-06	Per-tenant, model and serial NAT override delivered to the device	FR-35, FR-29	P1	✅ PASS	`unit` (precedence), `m6_server` (the override in the server's policy line) (Mac; CI build-test ×3, asan ×2)	unit.out, m6_server.out
TC-CFG-07	Profiles	FR-34	P2	✅ PASS	`unit`, `cli_config` (Mac; CI build-test ×3, asan ×2)	unit.out, cli_config.out
TC-CFG-08	Device clamp on advertised values	FR-35, NFR-06	P1	✅ PASS	`m2_peer` (Mac; CI build-test ×3, asan ×2)	m2_peer.out

Case	Title	Req	Pri	Result	Where it ran	Evidence
TC-CFG-09	Key-file permissions, and policy/artifact checks	FR-34, NFR-06	P1	✅ PASS	`unit`, `m3_register` (Mac; CI build-test ×3, asan ×2)	unit.out, m3_register.out
TC-CFG-10	Settings from the resolved decisions take effect (DESIGN §23)	FR-34, FR-35	P2	⚠ Partial	`m5_upgrade` (Mac; CI build-test ×3, asan ×2)	m5_upgrade.out — sub-case (f) runs (strict-IP resume, Linux CI); the other per-tenant/model/serial override sub-cases (a)-(e), (g) have no end-to-end check (the policy precedence is unit-tested, TC-UNIT-20)

6. Open and waived cases

The 15 open cases and the open parts of the partial ones that depend on tooling all need test tools that generate hostile or forged traffic: device hooks (`--bad-key`, `--replay-test`, `--reuse-nonce`, `--nonce`) and `peer_sim` modes (a rogue server, forged or replayed frames, floods). They were not built in this project and remain with the project owner (PRD R-15). They are counted as Open, not Waived: the owner accepted submitting without them, but they are not replaced by another test the way TC-SIM-04 is. The protection logic behind every one of them is implemented and was reviewed (DESIGN §18.3); README §18 lists, for each, which other tests reach that logic today. QA §12.5 is the register. The other partial cases are a platform or sub-case not run (each row says which).